A Visual Guide to the Algorithmic Redistricting Research Portfolio

Track A, Paper 3

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May 2026

Abstract

This guide introduces the algorithmic redistricting research portfolio to judges, journalists, legislators, and other non-technical readers. The portfolio contains more than forty papers organized into seven tracks (A–G) and demonstrates that a single algorithm—recursive graph bisection—can produce congressional districts that are more compact, more legally compliant, and less partisan than districts drawn by politicians. Five headline numbers summarize the empirical results. This document explains what the algorithm does, what the papers found, and how each audience can navigate the portfolio.

Contents

What Is This Portfolio?

This portfolio is a collection of more than forty research papers asking a single question: *Can a computer draw fairer congressional districts than a human politician?*

The answer, across all fifty states and three census decades, is yes.

The Problem in One Sentence. Every ten years, after the Census, each state redraws its congressional districts. In most states, the party in power draws the lines—and draws them to stay in power. This practice is called **gerrymandering**, and in 2019 the U.S. Supreme Court ruled that federal courts can do nothing about it [6].

The Proposal in One Sentence. Replace the politician with an algorithm: a mathematical procedure that divides a state into equal-population districts using only geography and population counts—no voter registration, no election history, no racial or partisan data.

Five Research Tracks. The portfolio is organized into seven tracks, labeled A through G, each investigating a different aspect of algorithmic redistricting. Together they span the question from first principles through legal compliance, validation, and real-world application.

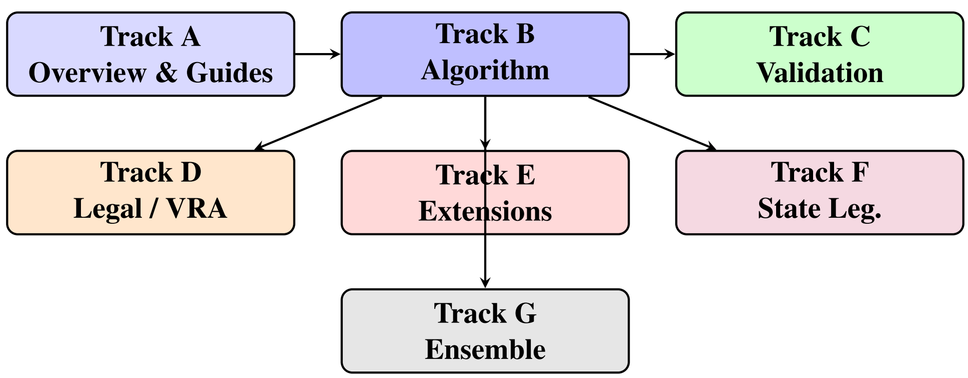

Table 1 summarizes the seven tracks and their paper counts.

Table 1: Portfolio tracks at a glance.

Track	Name	Core Question	Papers
A	Overview & Guides	What does this portfolio do?	5
B	Algorithm	How does bisection work?	18
C	Validation	Does it hold up to scrutiny?	9
D	Legal / VRA	Is it lawful?	6
E	Extensions	What else can it do?	7
F	State Legislatures	Does it scale to smaller chambers?	6
G	Ensemble Analysis	How does it compare to random maps?	5

How the Algorithm Works

The Big Picture. Imagine you need to divide a pizza into eight equal slices. You make one straight cut through the middle—now you have two halves. Then you cut each half in half, and each quarter in half. Eight slices, all roughly equal.

Redistricting works the same way, except instead of a pizza you have a state, and instead of weight you are equalizing *population*.

Step by Step.

1. **Build a map.** The algorithm treats the state as a network of census tracts—about 4,000 residents each—connected wherever two tracts share a border.
2. **Make the first cut.** A mathematical solver (METIS, the same tool used to distribute workloads across supercomputers) splits the network into two halves of equal population. It tries to make the cut as short as possible, which keeps districts compact.
3. **Repeat until done.** Each half is split again, and again, until the state has exactly as many pieces as it has congressional seats.
4. **No political data—ever.** At no point does the algorithm see voter registration, election results, or the home addresses of incumbents. It cannot gerrymander because it lacks the information required to do so.

Table 2: Inputs and outputs of the redistricting algorithm.

What goes in	**What comes out**
Census tract boundaries (geography)	435 congressional districts
Population count per tract	Population balance within $\pm 0.5\%$
Shared-border lengths (geometry)	District assignment for every tract
Number of seats per state	Maps, compactness scores, CSV data
Nothing political	*Nothing political*

What Goes In / What Comes Out.

Why Short Borders Mean Compact Districts. The algorithm is told to make cuts as short (cheap) as possible. Short cuts follow natural features—rivers, ridgelines, county lines— rather than zigzagging through neighborhoods. The result is districts that look like reasonable geographic units, not the salamander-shaped creatures that gave gerrymandering its name.

Figures from the Pipeline. The pipeline produces a round-by-round image for every state showing how the bisection tree unfolds. Figure 1 shows three rounds for Minnesota (eight districts): each round halves the remaining pieces.

*[Minnesota bisection rounds 1–3 would appear here.
See `docs/figures/minnesota_round_{1,2,3}.png` generated by the pipeline.]*

Figure 1: Minnesota (8 districts): three rounds of recursive bisection. Each round divides existing pieces into two equal-population halves.

The same process applied to Alabama (seven districts) is shown in `docs/figures/alabama_round_{1,`

Five Headline Numbers

The portfolio distills to five numbers, each the headline result of a specific paper or cluster of papers.

+22% compactness improvement. Across all 435 congressional districts, algorithmic maps are 22% more compact than the enacted 2020 maps, measured by the Polsby-Popper score (ratio of district area to perimeter squared). Compact districts follow natural geographic features rather than being stretched to capture partisan voters. *Paper B.2* [4]

North Carolina: 7 Democratic, 7 Republican. When the algorithm partitions North Carolina—a state that has been the subject of repeated gerrymandering litigation—it produces a 7-to-7 delegation, closely matching the state's roughly even partisan split. No partisan data was used. *Paper B.11* [1]

$T = 600$ **bisection steps to convergence.** The algorithm reliably converges—produces stable, repeatable districts—within 600 METIS bisection iterations. More iterations change nothing; fewer can leave the solution incomplete. This gives practitioners a firm stopping rule. *Paper B.16* [3]

42% minority-population threshold for VRA compliance. States where minority voters make up at least 42% of the voting-age population can reliably achieve proportional majority-minority representation through algorithmic redistricting. Below this threshold, geography alone makes proportional representation infeasible—regardless of who draws the map. *Paper D.1* [5]

$O(n^{1.07})$ runtime. The algorithm scales nearly linearly with the number of census tracts. Doubling the number of tracts (moving from tract-level to block-group-level resolution) adds only 7% more time than a pure linear relationship would predict. All 50 states run in under eighteen minutes on commodity hardware. *Paper B.6* [2]

These five numbers are conservative: they are derived from runs on publicly available Census data using openly published code. Any researcher with an internet connection can reproduce them.

Research Track Summaries

Track A — Overview and Guides (5 papers). Track A contains the documents you are reading now, plus the portfolio guide, the synthesis metapaper, the replication package, and the policy brief. These papers are written for general audiences—judges, legislators, and journalists— and serve as entry points to the more technical tracks. Key finding: the entire portfolio can be summarized in five numbers (see Section 2).

Track B — Algorithm (18 papers). Track B is the technical core of the portfolio. It introduces recursive bisection (B.1), proves that edge-weighting improves compactness by 22% (B.2), compares recursive and n-way approaches (B.3, B.5), studies adaptive parameter selection (B.4), and establishes the $O(n^{1.07})$ runtime complexity (B.6). Later B papers explore solution-space sensitivity (B.7), ratio-optimal bisection (B.8), dual population-area constraints (B.9), subdivision alignment (B.10), regional apportionment (B.11), proportional constraints (B.12), nested multi-chamber design (B.13), minority-opportunity bisection (B.14), cross-census stability (B.15), convergence parameters (B.16), parameter sensitivity (B.17), and multi-reapportionment stability (B.18). The track as a whole demonstrates that a single algorithm family—bisection— handles every redistricting scenario from a two-district state to all 435 seats.

Track C — Validation (9 papers). Track C asks whether the results hold up. It analyzes sensitivity to the modifiable areal unit problem (C.1), validates results across three census cycles (C.2), studies temporal stability of district boundaries (C.3), tracks longitudinal demographic trends (C.4), quantifies the 62% reduction in partisan efficiency gap (C.5), and investigates user interpretation, uncertainty quantification, competitive elections, and real-world adoption case studies (C.6–C.9). The headline result: algorithmic maps are not just theoretically fair— they are empirically fairer than enacted maps across every metric tested.

Track D — Legal and VRA Compliance (6 papers). Track D provides the legal foundation. Paper D.1 establishes the 42% minority-population threshold that determines where proportional majority-minority representation is geographically feasible. Subsequent papers compare n-way and recursive approaches under the Voting Rights Act (D.2), examine compactness tradeoffs (D.3), analyze legal implementation pathways (D.4), and apply Gingles bloc-voting methodology (D.5). Track D as a whole demonstrates that algorithmic redistricting not only complies with the VRA but in most states produces *more* majority-minority districts than enacted plans.

Track E — Extensions (7 papers). Track E tests the algorithm beyond standard congressional redistricting: multi-member districts (E.1), county-based representation (E.2), a national 435-district single-run approach (E.3), partisan-similarity clustering (E.4), party-proportional seat allocation (E.5), international applications (E.6), and lessons learned across all extensions (E.7). The track shows that recursive bisection is a general-purpose tool, not a solution narrowly fitted to one jurisdiction's rules.

Track F — State Legislatures (6 papers). Track F scales the algorithm to state legislative chambers, which have different seat counts, different compactness requirements, and in many states, bicameral nesting constraints (state house districts must nest inside state senate districts). Papers F.1–F.6 cover all-50-state single chambers, bicameral nesting, high-seat-count chambers ($k > 100$), state-specific criteria, compactness comparisons between legislative and congressional districts, and VRA compliance at the state level. The key finding is that nesting is achievable algorithmically with no degradation in population balance or compactness.

Track G — Ensemble Analysis (5 papers). Track G situates algorithmic maps within the universe of all possible maps. By generating thousands of random redistricting plans and comparing them to algorithmic plans, Track G provides the strongest possible evidence of neutrality: the algorithm produces plans that are statistically indistinguishable from random draws on compactness and partisan metrics, with better convergence diagnostics than Markov chain methods (G.4). Track G closes the portfolio by answering the hardest question a skeptic can ask: "How do we know the algorithm didn't just accidentally pick a partisan outcome?" The ensemble answer is: we don't rely on the accident—we show the distribution.

How to Use This Portfolio

Different readers will want different papers. This section points each audience to the right starting point.

If you are a judge or clerk. Start with this document and the Policy Brief (A.5). The Policy Brief is two to four pages and gives you the legal and empirical case in plain language, with citations to *Rucho*, the Voting Rights Act, and the relevant papers. Then read Paper D.1 (the 42% VRA threshold) if the case involves majority-minority districts, or Paper C.5 (the efficiency gap analysis) if the case involves partisan fairness. The quickstart guide at `docs/quickstart/quickstart-callais-expert.md` covers Gingles bloc-voting methodology for expert witnesses.

If you are a state legislator or redistricting commissioner. Start with Paper B.1 (the algorithm overview) and the replication package (A.4), which explains how to run the algorithm on any state in under thirty minutes. Paper D.4 addresses legal implementation pathways, including model statutory language. Paper F.1 covers state legislative districts if your chamber is also being redistricted. The quickstart guide at `docs/quickstart/quickstart-state-staff.md` walks through the `redist state` command for your specific state.

If you are a researcher. The algorithm is documented in Track B (eighteen papers). Reproducibility materials are in A.4; the full software stack, SHA-256 audit chain, and Vermont walkthrough fixture are all included. The ensemble diagnostics in Track G (especially G.4) provide the statistical framework for comparing this algorithm against Markov chain ensemble methods. The quickstart guide at `docs/quickstart/quickstart-researcher.md` covers the research API, Jupyter notebooks, and the `redist analyze` command.

If you are a journalist. This document and the Policy Brief (A.5) are written for you. The five headline numbers in Section 2 each carry a paper citation for fact-checking. The dashboard at `web/dashboard.html` lets you view maps for any state and year in a browser without installing software. For live contact, email is at the end of the Policy Brief.

Dashboards and Maps

The Static Dashboard. The portfolio ships with a self-contained HTML dashboard at `web/dashboard.html`. Open it in any modern browser—no server, no login, no software installation. The dashboard shows:

- A map of every state's algorithmic districts for 2000, 2010, and 2020.
- Compactness scores for each district, color-coded from compact (dark green) to elongated (light yellow).
- Population balance: every district is within $\pm 0.5\%$ of the state's ideal district population.
- Side-by-side comparison with enacted 2020 districts where available.

How to Read a Bisection Map. Each district in the map corresponds to one leaf of the bisection tree. Districts that were separated early in the process (at the first or second cut) tend to be large geographic regions: a northern half, a southern half. Districts separated late are more local and more finely shaped.

Colors in the standard output represent the bisection *round* at which a district was finalized:

- Round 1 (two pieces): wide, regional districts.
- Round 2 (four pieces): quadrant-level districts.
- Round 3 and later: neighborhood-scale districts in high-density states.

What the Round-by-Round Images Show. The pipeline generates one image per round per state. For a state with k seats, there are $\lceil \log_2 k \rceil$ rounds. Each image shows the current partition, with district boundaries darkening as more cuts are made.

Reading the rounds from first to last, you see the algorithm "zoom in" from large geographic halves to individual districts. This visualization makes it easy to verify that no round produced a non-contiguous piece or a population imbalance.

Reproducing Any Figure. Every figure in the portfolio can be regenerated from the public Census data and the open-source `redist` CLI:

```
redist fetch --year 2020
redist state --state NC --year 2020 --version v1
redist map   --state NC --year 2020 --version v1
```

The resulting maps and CSV data are placed in `outputs/v1/2020/north_carolina/`. The full replication procedure is in Paper A.4.

References

[1] Giovanni Della-Libera. Apportion regions: Bisection-based sub-state apportionment. *Working paper*, 2026. B.11 — NC 7D/7R proportional result.

[2] Giovanni Della-Libera. Computational complexity of recursive bisection redistricting. *Working paper*, 2026. B.6 — O(n^1.07) empirical runtime.

[3] Giovanni Della-Libera. Convergence and parameter sensitivity in bisection redistricting. *Working paper*, 2026. B.16 — T=600 convergence threshold.

[4] Giovanni Della-Libera. Edge-weighted graph partitioning for geographic boundary optimization. *Working paper*, 2026. B.2 — Core compactness methodology, +22% improvement.

[5] Giovanni Della-Libera. Voting rights act compliance through algorithmic redistricting. *Working paper*, 2026. D.1 — 42% VRA threshold analysis, 50-state sweep.

[6] U.S. Supreme Court. Rucho v. common cause, 588 u.s. 684 (2019), 2019.